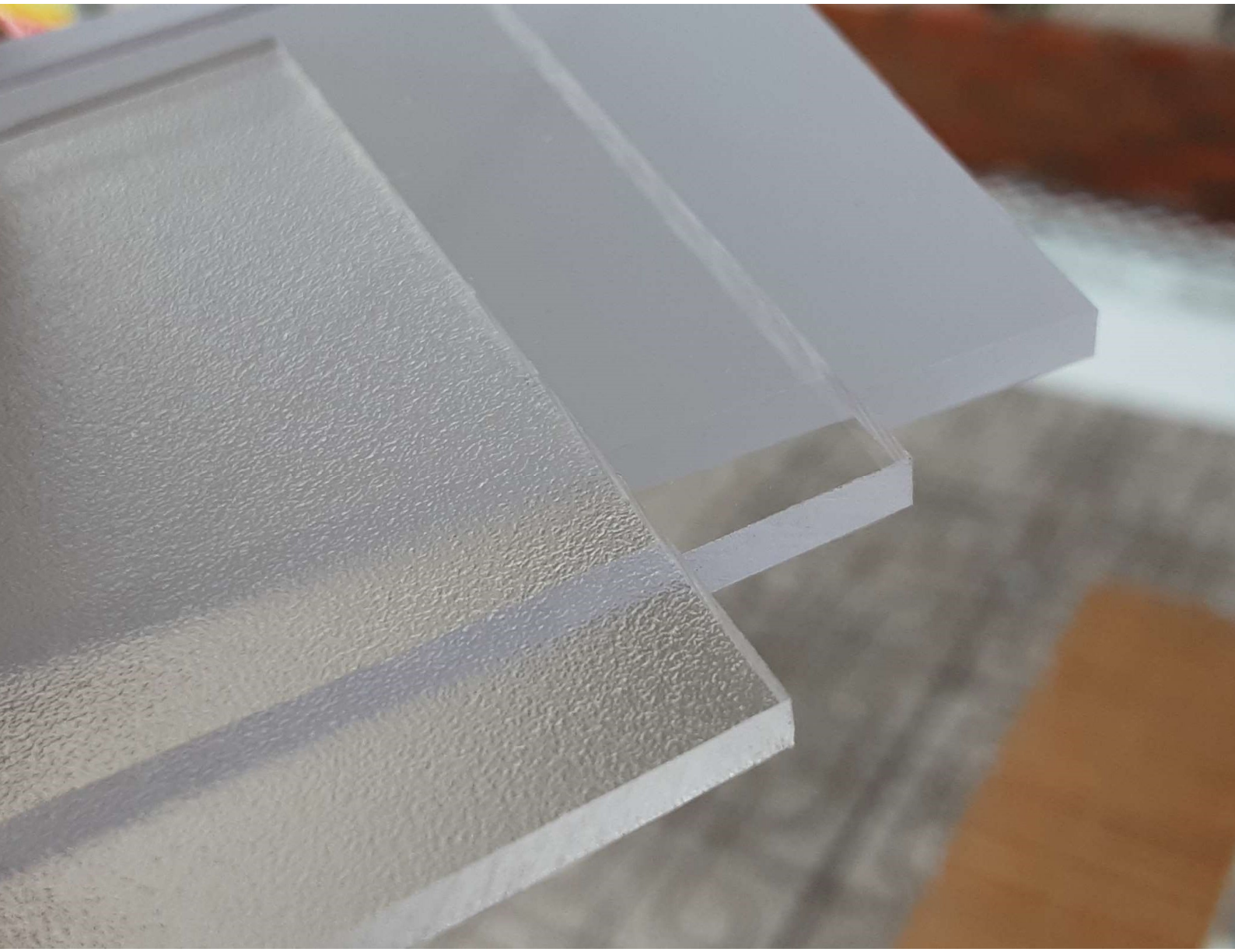
Coolmax® Polycarbonate Flat Solid Sheet

TECHNICAL DATA SHEET



SkyBuild+

Product Review

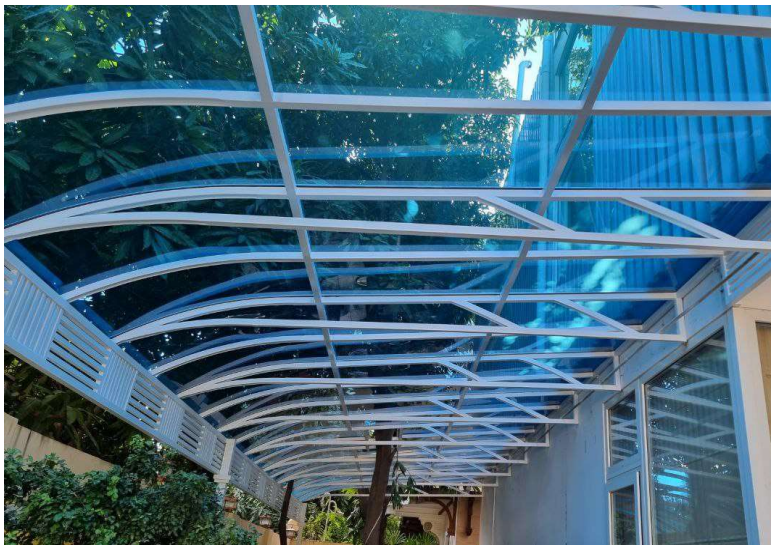
Coolmax® is polycarbonate flat solid sheet produced by the best PC resin and the latest advanced technology, making it virtually unbreakable and absolutely transparent.

Having superior clarity just like glass, very high impact strength, plentiful in color and easy to fabricate, **Coolmax®** is the best solution for glazing purpose which can be perfectly matched with your exterior and interior designs.

With **Coolmax®** roofing sheet, you can enjoy natural sunlight without concern about harmful UV rays because **Coolmax®** sheets block out up to 99% UV rays from sunlight.

Features

- Superior clarity and light transmission.
- Excellent impact resistance, 250 times stronger than glass and 20 times stronger than acrylic.
- Flexible, formable and easy to bend at normal temperature.
- Light weight (half of glass) helping eliminate the need of heavy support structures, saving cost in handling and installation.
- Easy to fabricate with conventional tools.
- UV protected and weather resistant.



Application



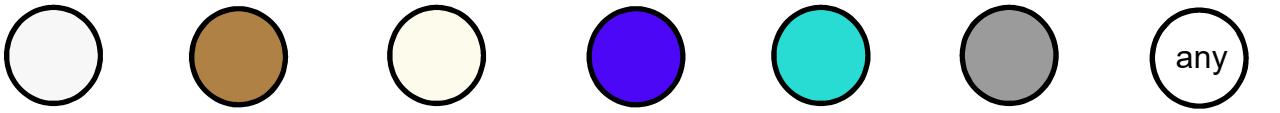
- Safety glazing.
- Skylight, roof, dome, canopy, terrace, awning, walkway, ..
- Car park, swimming pool, stadium, park, public utilities.
- Bus station, entrance to parking basement, metro station.
- Highway sound barrier.
- Greenhouse, drying domes for agriculture products, brick, charcoal...
- Signage, lightbox, decoration.
- Other applications in construction, industrial...

Standard Dimensions & Colors

Coolmax® polycarbonate flat solid sheet is available in the following dimensions and colors:

Thickness (mm)	2	3	5	10	customized
Width (mm)	1,220	1,520	1,800	2,100	customized
Length (mm)	20,000	30,000	6,000	2,440	customized
Weight (Kg/m²)	2.4	3.6	6	12	
Profile	Plain		Embossed		

Color



Clear | Bronze | Opal | Blue | Lake Blue | Grey | Customized

* Other dimensions and colors are available upon request, subject to minimum order quantity.

Chemical Resistance

Polycarbonate sheets are compatible with many chemicals and incompatible with a group of chemicals. For general guidance, please refer to below table:

Resistance	Non Resistance (damaged by)
Mineral acids	Acetone, ketones, ethers
Oxidizing and reducing agents	Aromatic and chlorinated hydrocarbons
Neutral and acid salt solutions	Aqueous or alcoholic alkaline solutions
Saturated, aliphatic and cycloaliphatic hydrocarbons and alcohols (except for methanol)	Ammonia gas and amines Methyl alcohol (methanol)

* Detailed list of Chemical Resistance is available upon request.

Typical Properties

Coolmax® Polycarbonate Flat Solid Sheet is produced by 100% virgin and finest PC resin (Covestro Makrolon® or Sabic), making it virtually unbreakable, having high clarity like glass, and resulting in good technical properties.

Following typical data is for thickness **3mm**:

Property	Unit	Result	Test Method
Physical			
Specific Gravity	g/cm³	1.2	ASTM D792
Water Absortion @ 24hr, 23 °C	%	0.06	ASTM D570
Mechanical			
Tensile Strength at yield	MPa	65.6	ASTM D638
Tensile Strength at break	MPa	66.7	ASTM D5420
Modulus of elasticity	Mpa	2460	ASTM D638
Elongation at break	%	102	ASTM D638
Flexural Strength	Mpa	118	ASTM D790
Hardness	Shore D	84	ASTM D2240
Thermal			
Services temperature	°C	- 20 to 120	-
Heat deflection Temperature, 1.82 Mpa	°C	135	ASTM D648
Ignition temperature, flash	°C	440	ASTM D1929
Coefficient of Thermal Expansion	mm/m/°C	0.065	ASTM D696
Flammability	-	V2	UL94
U-value	W/m².K	5.47	-
Optical			
Refractive Index	-	1.586	ASTM D542
Visible Light Transmission, Clear sheet	%	88	ASTM D1003
Acoustic			
Sound Reduction	dB	26	-

Winload and Framing Recommendation

Glazing frames for polycarbonate sheets can be of any form, popularly in rectangular. The ratio between the short dimension and long dimension of the frame is important to determine its ability to take the winload.

The below winload chart is recommended for 4 sides clamped flat installation:

Sheet thickness	Wind Load (kg/m²)	Maximum width (mm), based on Width : Length ratio	
		1:1	1:2
3mm	50	1050	800
	75	975	725
	100	875	650
	125	810	585
	150	760	525
	175	715	480
	200	685	450
5mm	50	1235	980
	75	1075	855
	100	980	775
	125	910	720
	150	855	680
	175	810	645
	200	775	615

Sheet thickness	Wind Load (kg/m²)	Maximum width (mm), based on Width : Length ratio	
		1:1	1:2
8mm	50	1975	1575
	75	1725	1375
	100	1565	1250
	125	1455	1160
	150	1375	1090
	175	1300	1035
	200	1250	950
10mm	50	2050	1600
	75	1950	1475
	100	1850	1400
	125	1730	1330
	150	1675	1275
	175	1625	1175
	200	1525	1100

Disclaimer: Information contained herein is intended only for technical persons, with any use thereof to be at their discretion. While we believe such information is reliable, Skybuild Plus or its agents shall have no liability for result obtained or damages resulting from such use. Content of this document is not construed as a warranty or guarantee by Skybuild Plus. Users should undertake sufficient verifications and testing to determine the suitability of the products for their particular purpose.